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Effective channel representations and composable security for passive-user Loop-Back quantum key distribution

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Abstract
This work develops effective channel representations and a composable-security framework for distributed passive-user Loop-Back quantum key distribution (QKD). Alice provides trusted preparation-and-measurement infrastructure at local service points, while two source-free and detector-free users, B_1 and B_2, apply independently selected polarization rotations $R(\pm\pi/8)$ to the same optical carrier. Two complementary derivations describe the honest operation. Direct averaging over the composed rotations gives the observable Loop-Back map, while an operationally equivalent representation on the Bennett–Brassard 1984 (BB84) preparation-and-measurement data is a balanced mixture of conjugate-basis dephasings, or an anisotropic Pauli channel with identity, X, and Z components and no Y component. Both recover the ideal conclusive-event probability $P_{\mathrm{conc}} = 1/4$. Security is established without restricting the adversarial action to this effective channel. Eve may implement an arbitrary coherent attack with quantum memory on the single untrusted interface between the two local rotations. In the single-photon sector, a Choi-operator identity yields the exact ideal relation $e_{1,\mathrm{p}} = 3e_{1,\mathrm{b}}/2$, together with a low-error robust angular bound $e_{1,\mathrm{p}} \leq \kappa_{\mathrm{rot}} e_{1,\mathrm{b}}$. Phase-randomized weak coherent pulses, decoy-state estimation and generalized entropy accumulation then give a finite-key composable bound. A common optical model compares asymptotic and finite-block rates and quantifies their convergence with block size. The analysis also identifies the trusted-interface condition required to exclude loss-assisted operation-discrimination attacks and specifies the hardware assumptions needed for optical implementation.

Keywords Quantum key distribution · Loop-Back QKD · Passive users · Quantum channels · Pauli channels · Composable security · Finite-key analysis

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1 Introduction

Quantum key distribution (QKD) provides a physical-layer mechanism for establishing secret correlations whose security does not rely on computational assumptions. Since the Bennett–Brassard 1984 (BB84) protocol (Bennett and Brassard 2014), most practical QKD architectures have followed a prepare-and-measure paradigm in which the communicating users require active quantum functionality, including state preparation, basis selection, synchronization, and, in many cases, single-photon detection. Although this model is well suited to laboratory demonstrations and backbone links, it remains demanding for lightweight, mobile, or densely deployed user nodes (Scarani et al. 2009).

A natural strategy to reduce this burden is to concentrate the active quantum functionality at a single station while restricting the remote side to simpler optical operations. Related asymmetric and two-way architectures include plug-and-play systems (Muller et al. 1997), semi-quantum protocols (Boyer et al. 2007, 2009), and deterministic two-way schemes (Lucamarini and Mancini 2005). Loop-Back QKD follows this principle: an active node prepares a quantum state, a remote node modifies or reflects the optical signal, and the state is returned for measurement (Lizama-Perez and López-Romero 2025; Lizama-Pérez and Morales-Calvo 2025). The relevant information is then associated with the accumulated round-trip transformation rather than with a direct receiver-side measurement. Bell-state and rotational extensions show that the same principle applies to entangled carriers while preserving source-free and detector-free user nodes (Lizama-Pérez 2026a, b, c).

This work considers a distributed passive-user version of this setting. A single active station, Alice, prepares and measures BB84 polarization states, whereas two remote users, B_1 and B_2, do not prepare or detect quantum states. Instead, each passive user applies a local polarization rotation to the same propagating quantum carrier. The operation observed by Alice is therefore the composed transformation

$$R_{\mathrm{eff}} = R_2 R_1. \tag{1}$$

For rotations selected from $\{R(+\pi/8), R(-\pi/8)\}$, this effective operation either cancels to the identity or produces a rotation toward the conjugate basis. Hence, conclusive events arise from the observable action of the composed passive subsystem, while the individual operations remain internal to the passive users.

The distributed passive-user mechanism admits two complementary channel derivations. The first follows the random composition of the physical rotations and produces the state transformation observed by Alice. The second expresses the same external statistics as a balanced mixture of dephasings in conjugate bases,

$$\mathcal{E}_{\mathrm{LB}}(\rho) = \frac{1}{2} D_Z(\rho) + \frac{1}{2} D_X(\rho). \tag{2}$$

and equivalently as an anisotropic Pauli channel with identity, X, and Z components and no Y component. These derivations describe the honest device and recover the ideal conclusive-event probability. They are not assumptions about the adversarial quantum channel.

The security analysis uses the physical topology of the intended service. Alice is trusted and may deploy preparation and detection modules at geographically distributed service

points. Each user acts locally inside such infrastructure. The trusted source is located immediately before B_1, the trusted receiver immediately after B_2, and Eve controls the single quantum interface between the rotations. Her action on that interface is arbitrary. It may include loss, adaptive operations and coherent quantum memory across rounds.

The contributions are therefore fourfold. First, the direct rotational derivation and the dephasing-channel derivation are shown to represent the same observable operation. Second, the dephasing form is reduced to a compact anisotropic Pauli channel. Third, an operator identity relates the phase-error and bit-error tests for every completely positive adversarial map allowed by the topology. Fourth, phase-randomized weak coherent pulses, decoy-state parameter estimation and the generalized entropy accumulation theorem are combined into a finite-key composable bound. The distinction between an honest effective model and an unrestricted adversarial map follows the standard separation between performance modeling and security analysis (Tomamichel and Leverrier 2017; Metger and Renner 2023).

2 Distributed passive-user Loop-Back configuration

We consider a minimal distributed Loop-Back configuration consisting of one active station, Alice, and two optically passive users, B_1 and B_2. Alice is the only node equipped with quantum state preparation and measurement capabilities. The passive users do not generate, measure, or store quantum states. They only apply local polarization rotations to the same propagating optical carrier.

The intended deployment is a local quantum service rather than a long-distance quantum link between the users. Alice operates trusted reader-like terminals at service points. The source and the input of B_1 belong to one trusted terminal boundary, while B_2 and the receiver belong to the next trusted boundary. The optical segment between the two rotations is untrusted. Figure 1 shows this causal partition.

A single protocol round follows the closed optical path

$$A \longrightarrow B_1 \longrightarrow B_2 \longrightarrow A. \tag{3}$$

Alice prepares a BB84 polarization state

$$|\psi_{b,k}\rangle \in \{|0_Z\rangle, |1_Z\rangle, |0_X\rangle, |1_X\rangle\}, \tag{4}$$

where $b \in \{Z, X\}$ denotes the preparation basis and $k \in \{0, 1\}$ denotes the encoded state within that basis. The state is sent through the two passive users and then returned to Alice, who measures it in the same basis b used for preparation.

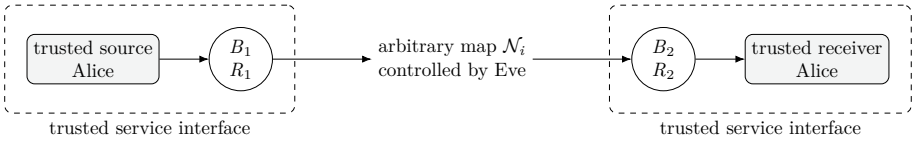


Fig. 1 Trusted-interface model. Alice supplies the source and receiver modules. Eve is unrestricted on the single intermediate quantum interface. The restriction is topological, not a restriction to a Pauli attack

Each passive user B_i, with $i \in \{1,2\}$, independently selects one of two polarization rotations, represented as qubit rotations in the corresponding Bloch-sphere plane,

$$R_i = R(\theta_i) = \exp(-i\theta_i Y) = \begin{pmatrix} \cos\theta_i & -\sin\theta_i \\ \sin\theta_i & \cos\theta_i \end{pmatrix}, \qquad \theta_i \in \left\{+\frac{\pi}{8}, -\frac{\pi}{8}\right\}. \tag{5}$$

Here, Y is the Pauli matrix generating rotations in the real polarization plane. Random unitary rotation is preferable to projective polarization filtering because it preserves the propagating carrier and does not introduce the basis-dependent attenuation of a polarizer. The equal-probability choices $R(+\theta)$ and $R(-\theta)$ are sufficient to reproduce the same averaged state transformation used below. The state returning to Alice is therefore

$$|\psi_{\text{out}}\rangle = R_2 R_1 |\psi_{b,k}\rangle. \tag{6}$$

Since the rotations act in the same polarization plane, their composition is additive:

$$R_2 R_1 = R(\theta_1 + \theta_2). \tag{7}$$

Thus, the effective rotation angle satisfies

$$\theta_{\text{eff}} = \theta_1 + \theta_2 \in \left\{-\frac{\pi}{4}, 0, +\frac{\pi}{4}\right\}. \tag{8}$$

There are two operational cases. If the users select opposite rotations, then

$$\theta_1 = -\theta_2, \qquad R_2 R_1 = R(0) = I, \tag{9}$$

and the state ideally returns unchanged to Alice. These rounds are inconclusive because Alice obtains the state originally prepared and cannot identify a nontrivial composed transformation.

If the users select the same rotation, then

$$\theta_1 = \theta_2, \qquad R_2 R_1 = R\left(\pm\frac{\pi}{4}\right). \tag{10}$$

In this case, the input state is rotated toward the conjugate basis. For example, for $|\psi_{b,k}\rangle = |0_Z\rangle$,

$$R\left(+\frac{\pi}{4}\right)|0_Z\rangle = |0_X\rangle, \qquad R\left(-\frac{\pi}{4}\right)|0_Z\rangle = |1_X\rangle. \tag{11}$$

A subsequent measurement by Alice in the original Z basis yields either $|0_Z\rangle$ or $|1_Z\rangle$ with equal probability. The observation of the state orthogonal to the one prepared is therefore a conclusive event, because it certifies that a nontrivial effective rotation was applied.

More generally, if Alice prepares $|\psi_{b,k}\rangle$ and measures in the same basis b, the conclusive-event rule is

$$\text{conclusive} \iff \Pi_m = |\psi^{\perp}_{b,k}\rangle\langle\psi^{\perp}_{b,k}|, \tag{12}$$

where Π_m is the projector associated with Alice's measurement outcome m, and $|\psi_{b,k}^{\perp}\rangle$ denotes the state orthogonal to $|\psi_{b,k}\rangle$ in basis b. Under ideal balanced choices,

$$\Pr(\theta_1 = \theta_2) = \frac{1}{2}, \tag{13}$$

and, conditioned on this event,

$$\Pr(m = \psi_{b,k}^{\perp} \mid \theta_1 = \theta_2) = \frac{1}{2}. \tag{14}$$

Therefore, the ideal conclusive-event probability is

$$\Pr(\text{conclusive}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}. \tag{15}$$

This probability is not introduced here as a final key rate. Rather, it characterizes the observable Loop-Back statistics produced by the distributed passive subsystem. We now show that these same statistics can be obtained by externally encapsulating B_1 and B_2 as a single effective Loop-Back node.

After the quantum transmission, Alice publicly announces only the indices of the conclusive rounds, over an authenticated classical channel. Ambiguous or inconclusive rounds are discarded. Alice does not reveal her prepared state or measurement outcome for the retained rounds. Hence, in the ideal case, the announcement only performs sifting. For each retained round, the condition of conclusiveness implies $\theta_1 = \theta_2$, so that each passive user can keep the bit associated with its own local rotation as a raw shared-key bit.

Alice's conclusive announcement reveals only that a nontrivial effective rotation occurred, equivalently $\theta_1 = \theta_2$, but it does not reveal whether the common local choice was $+\pi/8$ or $-\pi/8$. Hence, when the raw bit is assigned to the sign of the local rotation, B_1 and B_2 share the same raw-key bit, whereas Alice does not have access to its value.

3 Effective Loop-Back encapsulation

The distributed passive-user subsystem can be externally represented as a single effective Loop-Back node. From Alice's viewpoint, the internal decomposition into B_1 and B_2 is not directly accessible; only the global transformation applied to the returning quantum state is observed. We therefore define the effective passive subsystem

$$B^* = (B_1, B_2), \tag{16}$$

whose action is determined by

$$R_{\text{eff}} = R_2 R_1 = R(\theta_1 + \theta_2). \tag{17}$$

As shown above, the effective angle satisfies

$$\theta_{\text{eff}} \in \left\{-\frac{\pi}{4}, 0, +\frac{\pi}{4}\right\}. \tag{18}$$

The case $\theta_{\text{eff}} = 0$ leaves Alice's prepared state unchanged, whereas the cases $\theta_{\text{eff}} = \pm\pi/4$ map it toward the conjugate basis. Hence, the effective subsystem behaves as a node that either preserves Alice's preparation basis or induces a conjugate-basis action.

The first channel derivation follows the implemented random rotations directly. Since the zero angle occurs with probability 1/2 and each nonzero angle occurs with probability 1/4, the physical random-unitary channel is

$$\mathcal{E}_R(\rho) = \frac{1}{2}\rho + \frac{1}{4}R(+\pi/4)\rho R(+\pi/4)^\dagger + \frac{1}{4}R(-\pi/4)\rho R(-\pi/4)^\dagger. \tag{19}$$

Using $R(\pm\pi/4) = (I \mp iY)/\sqrt{2}$, the terms odd in Y cancel and

$$\mathcal{E}_R(\rho) = \frac{3}{4}\rho + \frac{1}{4}Y\rho Y. \tag{20}$$

For a Bloch vector (r_x, r_y, r_z), this map acts as

$$(r_x, r_y, r_z) \longmapsto \left(\frac{r_x}{2}, r_y, \frac{r_z}{2}\right). \tag{21}$$

Equation (20) is the channel generated by the optical implementation itself.

To make this relation explicit, suppose that Alice prepares $|0_Z\rangle$. In the identity case,

$$R(0)|0_Z\rangle = |0_Z\rangle. \tag{22}$$

In the nontrivial cases,

$$R\left(+\frac{\pi}{4}\right)|0_Z\rangle = |0_X\rangle, \qquad R\left(-\frac{\pi}{4}\right)|0_Z\rangle = |1_X\rangle. \tag{23}$$

Averaging over the two equally likely signs gives

$$\frac{1}{2}|0_X\rangle\langle 0_X| + \frac{1}{2}|1_X\rangle\langle 1_X| = \frac{I}{2}. \tag{24}$$

Thus, conditioned on a nontrivial effective rotation, the state returned to Alice is statistically indistinguishable from a state completely dephased in the conjugate basis.

The same reasoning applies to any BB84 input state

$$\rho_{b,k} = |\psi_{b,k}\rangle\langle\psi_{b,k}|, \qquad b \in \{Z, X\}, \quad k \in \{0, 1\}. \tag{25}$$

Let D_b denote complete dephasing in Alice's preparation basis b, and let $D_{\bar{b}}$ denote complete dephasing in the conjugate basis $\bar{b}$. Thus, if $b = Z$, then $D_b = D_Z$ and $D_{\bar{b}} = D_X$, whereas if $b = X$, then $D_b = D_X$ and $D_{\bar{b}} = D_Z$. Since $\rho_{b,k}$ is an eigenstate of basis b,

$$D_b(\rho_{b,k}) = \rho_{b,k}. \tag{26}$$

By contrast, complete dephasing in the conjugate basis gives

$$D_{\bar{b}}(\rho_{b,k}) = \frac{1}{2}\rho_{b,k} + \frac{1}{2}\rho^{\perp}_{b,k}, \tag{27}$$

where

$$\rho^{\perp}_{b,k} = |\psi^{\perp}_{b,k}\rangle\langle\psi^{\perp}_{b,k}|. \tag{28}$$

Under balanced passive-user choices, the effective Loop-Back action observed by Alice is therefore

$$\mathcal{E}_{\mathrm{LB}}(\rho_{b,k}) = \frac{1}{2}D_b(\rho_{b,k}) + \frac{1}{2}D_{\bar{b}}(\rho_{b,k}). \tag{29}$$

Substituting the two dephasing contributions yields

$$\begin{aligned}\mathcal{E}_{\mathrm{LB}}(\rho_{b,k}) &= \frac{1}{2}\rho_{b,k} + \frac{1}{2}\left(\frac{1}{2}\rho_{b,k} + \frac{1}{2}\rho^{\perp}_{b,k}\right)\\ &= \frac{3}{4}\rho_{b,k} + \frac{1}{4}\rho^{\perp}_{b,k}.\end{aligned} \tag{30}$$

Consequently, when Alice measures in her original preparation basis, the probability that the outcome is associated with the orthogonal projector is

$$\Pr\left(\Pi_m = \rho^{\perp}_{b,k}\right) = \frac{1}{4}, \tag{31}$$

which coincides with the conclusive-event probability obtained directly from the distributed transformation rule in (15).

Equation (30) shows that the distributed passive-user subsystem reproduces, at the level of Alice's observable statistics, the conclusive-event structure of an effective Loop-Back node. The distinction is internal: in the distributed implementation, the effective operation is not selected by a single remote party, but emerges from the private composition of two local transformations. Alice can identify whether a conclusive event occurred, but the internal decomposition of R_{eff} remains inaccessible from her measurement record alone.

The two channel derivations must be distinguished. The physical map $\mathcal{E}_R$ in (20) preserves the Y component, whereas the dephasing representative introduced above suppresses it. Nevertheless, every BB84 input used here has $r_y = 0$, and Alice measures only in the X or Z basis. Therefore,

$$\mathrm{Tr}[M\mathcal{E}_R(\rho_{b,k})] = \mathrm{Tr}[M\mathcal{E}_{\mathrm{LB}}(\rho_{b,k})] \tag{32}$$

for every protocol input $\rho_{b,k}$ and every protocol measurement element M. They are two completely positive and trace-preserving extensions compatible with the same accessible data, not identical maps on arbitrary qubit states. This non-uniqueness is expected because the protocol does not perform tomography along the Y axis.

4 Anisotropic Pauli-channel representation

The effective Loop-Back channel derived above was obtained from Alice's prepare-and-measure statistics on BB84 states. We now introduce a Pauli-channel representative for the externally encapsulated Loop-Back box $B^* = (B_1, B_2)$. This representative reproduces Alice's BB84 preparation-and-measurement statistics and provides a compact external model of the passive-user subsystem, without resolving the coherent internal sequence of transformations between B_1 and B_2. This representation makes explicit that the induced map is not an isotropic depolarizing channel, but a structured anisotropic Pauli channel associated with the two conjugate bases involved in the Loop-Back interaction.

Let

$$\rho = \frac{1}{2}\left(I + r_x X + r_y Y + r_z Z\right) \tag{33}$$

be an arbitrary qubit state, where X, Y, and Z are the Pauli matrices and

$$\mathrm{r} = (r_x, r_y, r_z) \tag{34}$$

is its Bloch vector. Complete dephasing in the Z basis removes the components transverse to the Z axis:

$$D_Z(\rho) = \frac{1}{2}\left(I + r_z Z\right). \tag{35}$$

Similarly, complete dephasing in the X basis removes the components transverse to the X axis:

$$D_X(\rho) = \frac{1}{2}\left(I + r_x X\right). \tag{36}$$

The effective Loop-Back operation observed by Alice is the balanced mixture

$$\mathcal{E}_{\mathrm{LB}}(\rho) = \frac{1}{2} D_Z(\rho) + \frac{1}{2} D_X(\rho). \tag{37}$$

Substituting the expressions of D_Z and D_X into (37), we obtain

$$\begin{aligned} \mathcal{E}_{\mathrm{LB}}(\rho) &= \frac{1}{2}\left[\frac{1}{2}\left(I + r_z Z\right)\right] + \frac{1}{2}\left[\frac{1}{2}\left(I + r_x X\right)\right] \\ &= \frac{1}{2} I + \frac{1}{4} r_x X + \frac{1}{4} r_z Z. \end{aligned} \tag{38}$$

Thus, in the Bloch representation, the effective Loop-Back channel acts as

$$(r_x, r_y, r_z) \longmapsto \left(\frac{r_x}{2},\, 0,\, \frac{r_z}{2}\right). \tag{39}$$

The Y-component is completely suppressed, whereas the X- and Z-components are contracted by a factor 1/2. This already shows that the channel is anisotropic.

Proposition 1 *The balanced effective Loop-Back channel*

$$\mathcal{E}_{\mathrm{LB}}(\rho) = \frac{1}{2}D_Z(\rho) + \frac{1}{2}D_X(\rho) \tag{40}$$

is equivalent to the anisotropic Pauli channel

$$\mathcal{E}_{\mathrm{LB}}(\rho) = \frac{1}{2}\rho + \frac{1}{4}X\rho X + \frac{1}{4}Z\rho Z. \tag{41}$$

Proof Consider the general Pauli channel

$$\mathcal{P}(\rho) = p_I\rho + p_X X\rho X + p_Y Y\rho Y + p_Z Z\rho Z. \tag{42}$$

Set

$$p_I = \frac{1}{2}, \qquad p_X = \frac{1}{4}, \qquad p_Y = 0, \qquad p_Z = \frac{1}{4}. \tag{43}$$

These coefficients satisfy $p_I + p_X + p_Y + p_Z = 1$, and the map is completely positive and trace preserving because it is a convex mixture of unitary Pauli conjugations. Then

$$\mathcal{P}(\rho) = \frac{1}{2}\rho + \frac{1}{4}X\rho X + \frac{1}{4}Z\rho Z. \tag{44}$$

Using

$$X\rho X = \frac{1}{2}\left(I + r_x X - r_y Y - r_z Z\right), \tag{45}$$

$$Z\rho Z = \frac{1}{2}\left(I - r_x X - r_y Y + r_z Z\right), \tag{46}$$

and substituting into (44), we obtain

$$\begin{aligned}\mathcal{P}(\rho) &= \frac{1}{2}\left[\frac{1}{2}\left(I + r_x X + r_y Y + r_z Z\right)\right] \\ &\quad + \frac{1}{4}\left[\frac{1}{2}\left(I + r_x X - r_y Y - r_z Z\right)\right] \\ &\quad + \frac{1}{4}\left[\frac{1}{2}\left(I - r_x X - r_y Y + r_z Z\right)\right] \\ &= \frac{1}{2}I + \frac{1}{4}r_x X + \frac{1}{4}r_z Z.\end{aligned} \tag{47}$$

This coincides with the Bloch action in (38). Therefore,

$$\mathcal{P}(\rho) = \mathcal{E}_{\mathrm{LB}}(\rho), \tag{48}$$

which proves the claim. □

Equation (41), together with Fig. 2, provides a compact channel-level representative of the passive-user Loop-Back statistics. The identity component represents the part of the process compatible with Alice's preparation basis, whereas the X and Z components encode basis-dependent flips associated with the conjugate-basis branch. The absence of a Y component characterizes this representative. It does not imply that the protocol data uniquely determine the action on an arbitrary Y-polarized input.

This distinction is operationally relevant. In an isotropic depolarizing channel, the three Pauli components appear symmetrically. Here, by contrast, the representative preserves the special role of the X and Z bases used by the Loop-Back interaction. It describes intrinsic statistics induced by the distributed passive transformation structure. Neither representative is used to restrict Eve in the security proof below.

5 Composable security against an unrestricted intermediate attack

The effective channels in Sects. 3 and 4 model honest statistics. Security is instead proved for an arbitrary adversarial operation on the intermediate interface. The assumptions and classical processing are stated before the entropy bound so that the claim has an explicit physical scope.

5.1 Trust and device assumptions

Alice is trusted. Her source, preparation register, intensity setting, receiver and internal records are not controlled by Eve. Each user generates a fresh, independent and uniform bit and implements a characterized rotation. Eve controls the optical system between the two rotations. In round , her most general action is a completely positive and trace-preserving map

$$\mathcal{N}_i : Q_i E_{i-1} \longrightarrow Q_i' E_i, \tag{49}$$

where Q_i is the optical system and E_i is Eve's updated quantum memory. Loss is included as an output flag. The map may depend on all previous public information, which covers adaptive coherent attacks.

The logical proof requires a qubit interface with a classical loss flag. For an optical receiver, let $M_j = I_2 - \rho_j$ denote the logical acceptance effect for preparation ρ_j. The full

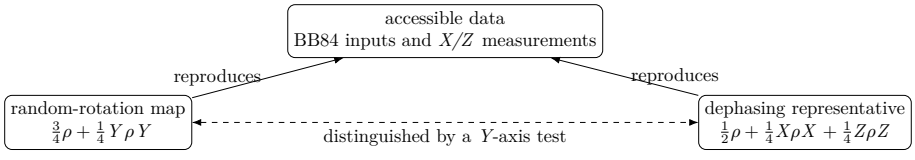


Fig. 2 Two channel representatives compatible with the protocol data. They agree on the operational X-Z slice but differ on an unmeasured Y component

optical effects $F_{j,y}^{\mathrm{opt}}$, including the second user rotation, must admit a common completely positive and trace-preserving reduction $\mathcal{S}$, independent of j, y, such that

$$F_{j,y}^{\mathrm{opt}} = \mathcal{S}^{*}\big(R_2(y)^{\dagger} M_j R_2(y)\big)\,. \tag{50}$$

Here $\mathcal{S}^{*}$ is the adjoint map. The complementary outcomes and loss must also be reproduced. A reduction covariant under the relevant rotations is sufficient. The existence of a squashing model for a different measurement family does not establish this condition (Beaudry et al. 2008). Double clicks require a predetermined basis-independent rule. The optical extension is conditional on this reduction or an explicit optical proof, including possible multiphoton inputs injected by Eve.

The classical channel is authenticated. Alice randomizes the global phase of weak coherent pulses, and the intensities are indistinguishable after conditioning on photon number (Lo and Preskill 2007; Ma et al. 2005). Each rotation magnitude is common to its two signs and, conditioned on the preceding history, is independent of the current preparation and intensity. Past private key records do not control subsequent optical rounds. Each round completes its optical interaction and receiver decision before the next round is processed, while Eve's memory may persist. Secrecy is claimed against Eve, with Alice inside the trusted domain.

5.2 Protocol used in the proof

Alice chooses an intensity $\lambda_i \in \{\mu, \nu, 0\}$, where $\mu > \nu > 0$, and prepares one of the four BB84 states uniformly. User B_1 chooses $x_i \in \{0, 1\}$ and applies

$$R_1(x_i) = \exp[-i(-1)^{x_i} a_1 Y]. \tag{51}$$

Eve applies (49). User B_2 independently chooses y_i and applies the analogous rotation $R_2(y_i)$. Alice measures in her preparation basis and announces the acceptance flag F_i. The value $F_i = 1$ denotes the outcome orthogonal to her prepared state. Vacuum and no-click outcomes follow the characterized receiver rule.

For each accepted signal-intensity round, a fresh independent test flag T_i is generated with $\Pr(T_i = 1) = \tau$, where $0 \leq \tau < 1$ is fixed before acquisition. Equivalently, the flag may be generated earlier and kept private until the optical interaction ends. The users disclose their bits in signal test rounds and in accepted weak-decoy rounds. Accepted signal rounds with $T_i = 0$ form the raw key. The bits are x_i for B_1 and y_i for B_2, with error event $x_i \neq y_i$. All disclosures and selection flags enter the public transcript.

The emitted block size, statistical acceptance region, final key length and classical communication budget are predetermined. Raw keys undergo reconciliation, verification and privacy amplification with independent two-universal hash seeds. A two-universal family has collision probability at most 2^{-t} for distinct inputs when its output has t bits. The protocol aborts if reconciliation exceeds its fixed communication budget. The independent test flags specify the sampling rule used in the proof.

5.3 Composable definition

Let K_1 and K_2 be the final keys and let Ω_{fin} denote non-abort. Correctness with parameter ε_{cor} means

$$\Pr[K_1 \neq K_2 \wedge \Omega_{\text{fin}}] \leq \varepsilon_{\text{cor}}. \tag{52}$$

Secrecy with parameter ε_{sec} means

$$\frac{1}{2}\Pr[\Omega_{\text{fin}}] \left\| \rho_{K_1 EI|\Omega_{\text{fin}}} - \tau_{K_1} \otimes \rho_{EI|\Omega_{\text{fin}}} \right\|_1 \leq \varepsilon_{\text{sec}}, \tag{53}$$

where I is the public transcript and τ_{K_1} is a uniform key independent of Eve. The two conditions give $\varepsilon_{\text{cor}} + \varepsilon_{\text{sec}}$ composable security (Renner 2005; Portmann and Renner 2022).

5.4 Virtual choices and observable errors

For the single-photon sector, the random choices are purified into qubit registers S_1 and S_2 . Before their final computational-basis measurements, both registers are prepared virtually in $|+\rangle$ and control the corresponding rotations. The ideal accepted target is

$$|\Phi^-\rangle_{S_1 S_2} = \frac{|00\rangle - |11\rangle}{\sqrt{2}}. \tag{54}$$

Define the bit-error and phase-error projectors

$$P_{\text{b}} = \frac{I - Z \otimes Z}{2}, \qquad P_{\text{p}} = \frac{I + X \otimes X}{2}. \tag{55}$$

The sign in P_{p} follows from the negative relative phase of $|\Phi^-\rangle$. Let $q_{1,\text{b}}$ and $q_{1,\text{p}}$ be the non-normalized accepted bit and phase errors in the single-photon sector. Dividing by its acceptance probability gives $e_{1,\text{b}}$ and $e_{1,\text{p}}$.

Lemma 2 *(Choi-test identity) For ideal angles $a_1 = a_2 = \pi/8$, uniformly random user bits and the uniform BB84 ensemble,*

$$q_{1,\text{p}} = \frac{3}{2} q_{1,\text{b}}, \qquad e_{1,\text{p}} = \frac{3}{2} e_{1,\text{b}}. \tag{56}$$

The identities hold for every completely positive map on the intermediate interface, including trace-nonincreasing branches of an attack with loss.

The proof is given in Appendix A. It constructs operators Γ_{b} and Γ_{p} that test the Choi matrix of the adversarial map and verifies the operator identity $\Gamma_{\text{p}} = 3\Gamma_{\text{b}}/2$. No Pauli-channel assumption is made.

For imperfect rotations in the regime $e_{1,\text{b}} \leq 1/2$, suppose that $|a_i - \pi/8| \leq \Delta_i$ and the allowed intervals remain inside $(0, \pi/4)$. Appendix B gives

$$e_{1,\mathrm{p}} \le \kappa_{\mathrm{rot}} e_{1,\mathrm{b}}, \qquad \kappa_{\mathrm{rot}} = 1 + \cos\Big(\frac{\pi}{4} - 2\Delta_1\Big)\cos\Big(\frac{\pi}{4} - 2\Delta_2\Big). \tag{57}$$

For larger bit errors, the entropy contribution is set to zero rather than extending this inequality beyond its derivation. For $\Delta_1 = \Delta_2 = 1^\circ$, one obtains $\kappa_{\mathrm{rot}} = 1.5348782369$.

5.5 Entropy and decoy-state reduction

The smooth-entropy uncertainty relation with quantum side information converts the phase-error bound into secrecy (Berta et al. 2010; Tomamichel and Renner 2011; Tomamichel et al. 2012). For an accepted single-photon key round not disclosed for testing,

$$H(K \mid E, I_{\mathrm{pre}}, F, m = 1) \ge 1 - h_2\Big(\min\Big\{\frac{1}{2}, \kappa_{\mathrm{rot}} e_{1,\mathrm{b}}\Big\}\Big), \tag{58}$$

Here m denotes emitted photon number, I_{pre} contains the current and preceding public information before reconciliation, $h_2(u) = -u\log_2 u - (1-u)\log_2(1-u)$ is binary entropy with its continuous endpoint values, and K is the computational-basis outcome of S_1. The bound includes conditioning on any purifying reference of Eve's input memory. It is averaged over the accepted preparation ensemble. When $\kappa_{\mathrm{rot}} e_{1,\mathrm{b}} \ge 1/2$, nonnegativity of classical conditional entropy supplies the zero lower bound.

For a phase-randomized pulse of mean photon number λ,

$$\rho_\lambda = \sum_{m=0}^{\infty} e^{-\lambda}\frac{\lambda^m}{m!}\rho_m. \tag{59}$$

Let $Q_{\lambda,F}$ be the accepted gain and $T_{\lambda,F}$ the accepted bit-error gain. Let $Y_{1,F}$ and $Z_{1,\mathrm{b}} = Y_{1,F} e_{1,\mathrm{b}}$ be the corresponding single-photon quantities. Vacuum plus one weak decoy give (Ma et al. 2005; Lim et al. 2014)

$$Y_{1,F}^{L} = \frac{\mu}{\mu\nu - \nu^2}\left[e^{\nu}Q_{\nu,F} - \frac{\nu^2}{\mu^2}e^{\mu}Q_{\mu,F} - \frac{\mu^2 - \nu^2}{\mu^2}Q_{0,F}\right], \tag{60}$$

$$Z_{1,\mathrm{b}}^{U} = \frac{e^{\nu}T_{\nu,F} - Q_{0,F}/2}{\nu}. \tag{61}$$

Superscripts L and U denote lower and upper bounds. The vacuum error is $1/2$ because B_1's fresh bit is independent of any injected optical state when Alice emits vacuum.

The asymptotic rate per signal-intensity emission, including the independent test fraction, is

$$R_\mu \ge (1-\tau)\left\{\mu e^{-\mu} Y_{1,F}^{L}\left[1 - h_2\left(\min\left\{\frac{1}{2}, \kappa_{\mathrm{rot}}\frac{Z_{1,\mathrm{b}}^{U}}{Y_{1,F}^{L}}\right\}\right)\right] - Q_{\mu,F} f_{\mathrm{EC}} h_2(E_{\mu,F})\right\}, \tag{62}$$

where $E_{\mu,F} = T_{\mu,F}/Q_{\mu,F}$ and $f_{\mathrm{EC}} \ge 1$ is the inefficiency of error correction.

5.6 Finite-key theorem

The affine min-tradeoff function in Appendix C lower-bounds the entropy of each round conditioned on Eve's updated memory, the public transcript and a purifying reference of the input. This permits application of the generalized entropy accumulation theorem (GEAT) (Metger et al. 2024; Metger and Renner 2023; Kamin et al. 2025). Let n be the emitted block size. For a predetermined region $\mathcal{S}_{\mathrm{acc}}$ of empirical score distributions, define

$$\Omega_{\mathrm{PE}} = \{\hat{q} \in \mathcal{S}_{\mathrm{acc}}\}, \qquad h = \min_{q \in \mathcal{S}_{\mathrm{acc}}} f(q). \tag{63}$$

The statistical event Ω_{PE} precedes reconciliation. The final non-abort event Ω_{fin} may be smaller.

Round outputs a padded binary register K_i, equal to x_i only if $\lambda_i = \mu, F_i = 1, T_i = 0$, and equal to a fixed zero otherwise. The useful-position pattern is public, so deleting fixed positions is reversible given that pattern and preserves the conditional min-entropy. The sequential maps retain all public records in Eve's side information. Private records are inert during later optical rounds. Appendix C verifies non-signalling and reconstruction of the public score sequence.

Theorem 3 *(Fixed-length composable security) Under the trust, device and sequential-processing assumptions above, fix $1 < \alpha < 3/2$, smoothing parameter $\varepsilon_{\mathrm{s}} \in (0,1)$, entropy-accumulation threshold $\varepsilon_{\mathrm{EA}} \in (0,1)$, error-verification parameter $\varepsilon_{\mathrm{EV}} \in (0,1)$ and privacy-amplification parameter $\varepsilon_{\mathrm{PA}} \in (0,1)$. Fix a reconciliation transcript alphabet of at most $2^{L_{\mathrm{EC}}}$ values, conditioned on previous public records and independent seeds. A predetermined positive integer key length ℓ satisfying*

$$\begin{aligned} \ell \leq & nh - \Delta_{\mathrm{GEAT}}(n, \alpha, f, \varepsilon_{\mathrm{s}}, \varepsilon_{\mathrm{EA}}) - L_{\mathrm{EC}} \\ & - \left\lceil \log_2 \frac{1}{\varepsilon_{\mathrm{EV}}} \right\rceil - 1 - 2\log_2 \frac{1}{2\varepsilon_{\mathrm{PA}}} \end{aligned} \tag{64}$$

produces an $\varepsilon_{\mathrm{EV}}$-correct and $\varepsilon_{\mathrm{sec}}$-secret key, where

$$\varepsilon_{\mathrm{sec}} = \varepsilon_{\mathrm{EA}} + 2\varepsilon_{\mathrm{s}} + \varepsilon_{\mathrm{PA}}. \tag{65}$$

The verification hash leaks the displayed ceiling term. The additional bit conservatively accounts for its public pass/fail announcement. All other data-dependent classical messages and branch labels are included in L_{EC}. With ideal authentication, the total composable parameter is $\varepsilon_{\mathrm{EV}} + \varepsilon_{\mathrm{sec}}$.

The proof and explicit correction are given in Appendix C. This construction uses a fixed empirical acceptance region and does not require an additional confidence-interval failure parameter. A separate estimation procedure or an adaptive output length requires its own reduction. Authentication failure, if nonzero, adds its declared parameter to the composable budget. Completeness concerns honest abort and is treated separately.

The theorem includes collective attacks and adaptive coherent attacks on the stated sequential interface. Parameters and the tangent defining f are fixed before acquisition. If the key bound is nonpositive, no key is generated. The protocol also aborts on failure of the statistical tests, exhaustion of the reconciliation budget or failed verification. The accep-

tance region may enforce conservative yield and error thresholds through empirical scores without identifying those scores with exact channel probabilities.

5.7 Intermediate-state leakage and boundary of the claim

For a BB84 input $|\psi\rangle$, the two intermediate states after B_1 obey

$$|\langle\psi|R(+\pi/8)^{\dagger}R(-\pi/8)|\psi\rangle| = \frac{1}{\sqrt{2}}. \tag{66}$$

They are nonorthogonal. Within the single intermediate interface, Lemma 2 quantitatively converts every adversarial information-gathering operation into an observable error constraint and the entropy bound (58) limits Eve's retained quantum information.

The topology cannot be weakened without changing the claim. If Eve can inject a chosen probe before one rotator and recover it after that same rotator, optimal unambiguous discrimination of $R(+\pi/8)$ and $R(-\pi/8)$ succeeds with probability

$$p_{\mathrm{UD}} = 1 - \frac{1}{\sqrt{2}} \simeq 0.292893. \tag{67}$$

Eve may conceal inconclusive discrimination outcomes as optical loss. The trusted source immediately before B_1 and trusted receiver immediately after B_2 therefore form security assumptions, not implementation conveniences. Optical isolation, input-power monitoring and spectral filtering are required to enforce this boundary (Gisin et al. 2006; Lucamarini et al. 2015).

The earlier lossless intercept–measure–resend partition remains useful as an illustrative attack calculation. Its probabilities 1/4, 3/16 and 9/16 describe useful, erroneous and inconclusive sectors for that attack only. The value 1/4 is the ideal accepted-event probability, not a universal secret-key rate. The composable rate is determined by (64).

6 Optical implementation and parameter-estimation guidance

Random rotations are more suitable than projective polarizers for the passive terminals. A polarization rotator implements a unitary transformation and preserves the carrier apart from insertion loss. A polarizer performs a nonunitary filter, introduces state-dependent attenuation and changes the loss model. The required $\pm\pi/8$ settings may be implemented with calibrated wave plates or electro-optic polarization modulators. Their angular uncertainty enters security through κ_{rot}, rather than being absorbed into an unspecified visibility parameter.

For a practical weak-coherent implementation, the overall transmittance η_{sys} contains propagation loss, component insertion loss and detector efficiency. Dark counts contribute to $Q_{0,F}$ and approach a bit-error fraction of 1/2. Detector efficiency mismatch and basis-dependent response are not covered by a scalar efficiency. They must either be included in a validated squashing model or treated by an extended detector-flaw proof (Makarov et al. 2006). Table 1 summarizes the required treatment.

Table 1 Implementation parameters and their role in the security model

Parameter	Observable effect	Required treatment
Propagation and insertion loss	Reduced accepted gains	Included in $Q_{\lambda,F}$, but not trusted as evidence of secrecy
Detector efficiency	Reduced click probability	Included in the characterized receiver or squashing map
Dark-count yield	Vacuum clicks and random errors	Estimated through $Q_{0,F}$
Double clicks	Ambiguous logical outcome	Predetermined basis-independent assignment
Angular error Δ_i	Enlarged phase-error bound	Included through κ_{rot}
Intensity fluctuation	Biased photon-number bounds	Calibrated intervals or a fluctuation-aware decoy analysis
Back-reflection leakage	Possible disclosure of user settings	Isolation, filtering, monitoring and a certified photon budget
Spectral or temporal labels	Violation of decoy indistinguishability	Characterization or explicit side-channel model

Table 2 Parameters shared by the asymptotic and finite-block evaluations

Parameter	Value
Signal and weak-decoy intensities (μ, ν)	$(0.5, 0.1)$
Intensity probabilities (p_μ, p_ν, p_0)	$(0.80, 0.15, 0.05)$
Signal test probability τ	0.05
Detector efficiency η_{d}	0.20
Background parameter Y_0, optical noise u	10^{-7}, 0.02
Angular uncertainty $\Delta_1 = \Delta_2$	1°
Reconciliation inefficiency f_{EC}	1.10
$\varepsilon_{\mathrm{s}}, \varepsilon_{\mathrm{EA}}, \varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}}$	10^{-10} each
Honest score-test failure target $\varepsilon_{\mathrm{score}}$	10^{-3}

6.1 Common optical model

For the illustrative honest model, let L be propagation and insertion loss in decibels, η_{d} detector efficiency, $\eta = \eta_{\mathrm{d}} 10^{-L/10}$, u an optical noise parameter, and Y_0 a background parameter. The accepted gain and error gain are defined by

$$Q_{\lambda,F} = (1 - e^{-\eta\lambda})\frac{1+u}{4} + e^{-\eta\lambda}\frac{Y_0}{2}, \tag{68}$$

$$T_{\lambda,F} = (1 - e^{-\eta\lambda})\frac{u}{4} + e^{-\eta\lambda}\frac{Y_0}{4}. \tag{69}$$

These equations specify reference performance only and are not restrictions on the attack map.

Table 2 specifies the common parameters used in the asymptotic and finite-block evaluations. All rates are normalized per total emitted pulse, including signal, weak-decoy and

vacuum choices. The accepted vacuum gain is $Q_{0,F} = Y_0/2 = 5 \times 10^{-8}$. Detector efficiency is included separately from the loss variable L. The model is a reference operating point used to select a protocol design before acquisition. It is not an inferred restriction on Eve.

6.2 Predetermined finite-block designs

For each nominal loss L, the common model gives a reference score distribution q^{ref} through Appendix C. The tangent is selected at $e_\star = e^U_{1,\mathrm{b}}(q^{\mathrm{ref}})$, in the interior region $\kappa_{\mathrm{rot}} e_\star < 1/2$. This choice makes the affine entropy at the reference point equal to the decoy lower bound on the privacy contribution,

$$f(q^{\mathrm{ref}}) = p_\mu(1-\tau)\mu e^{-\mu} Y^L_{1,F}\left[1 - h_2(\kappa_{\mathrm{rot}} e_\star)\right]. \tag{70}$$

The coefficients r_c and range R_f are defined in Appendix C. For each block size n, fix

$$m_n = R_f \sqrt{\frac{\ln(1/\varepsilon_{\mathrm{score}})}{2n}}, \qquad h_n = f(q^{\mathrm{ref}}) - m_n, \tag{71}$$

$$\mathcal{S}_{\mathrm{acc}} = \{q : f(q) \geq h_n\}. \tag{72}$$

Under independent honest emissions, the bounded-score exponential-moment argument in Appendix D gives $\Pr[f(\hat{q}) < h_n] \leq \varepsilon_{\mathrm{score}}$. This is a bound on failure of the score test under the honest model. It is not a new term in the adversarial secrecy budget and does not bound reconciliation failure.

Set the reconciliation budget before the block to

$$L_{\mathrm{EC}} = \lceil n c_{\mathrm{EC}} \rceil, \quad c_{\mathrm{EC}} = p_\mu(1-\tau) f_{\mathrm{EC}} Q_{\mu,F} h_2(E_{\mu,F}), \tag{73}$$

where $E_{\mu,F} = T_{\mu,F}/Q_{\mu,F}$. This uses a conventional reconciliation cost as a fixed alphabet budget, not an observed syndrome length. The protocol aborts if its implementation cannot complete reconciliation within this budget. No particular code or code-success probability is assumed by the plotted secrecy bound.

For each design pair (L, n), choose α to minimize the explicit correction in Appendix C, before collecting data. Substitute $h = h_n$ and (73) into (64). Its positive part divided by n defines the plotted rate r_n. Integer key lengths are rounded down in an instantiated protocol, a change of less than $1/n$ in the plotted rate. The nominal loss, tangent, acceptance threshold, budget and output length are all predetermined. Different points on a curve represent different fixed protocol designs, not adjustments made after observing a block. For ideal authentication the total security parameter is 5×10^{-10}.

6.3 Comparison and finite-size scaling

Figure 3 compares the asymptotic rate with the finite-block rates under the same optical model. The asymptotic positive-rate threshold is approximately 46.68 decibels (dB) of propagation and insertion loss. Table 3 shows that the conservative finite-block thresholds are 5.32, 25.25 and 43.12 dB for $n = 10^{12}, 10^{16}, 10^{20}$, respectively. A zero plotted rate means that

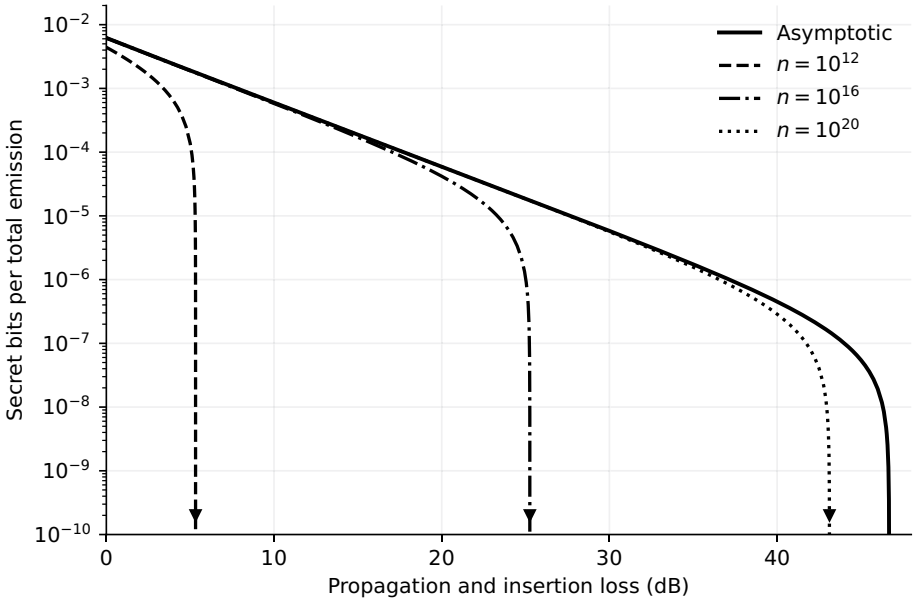


Fig. 3 Asymptotic and finite-block secret bits per total emission under the common model in Table 2. Only positive rates are shown on the logarithmic axis. Bottom triangles locate the zero crossings of the finite-block bounds, not nonzero rates at those positions. Each finite curve uses the predetermined design of Sect. 6.2. Rates are conditional on successful completion of the stated tests and classical processing

Table 3 Positive-rate thresholds of the common-model evaluation

The loss excludes the separately included detector efficiency. Rates are per total emission

Design	Positive-rate threshold (dB)	Rate at zero loss
Asymptotic	46.68	0.0062
$n = 10^{12}$	5.32	0.00442
$n = 10^{16}$	25.25	0.00618
$n = 10^{20}$	43.12	0.0062

this bound and the selected parameters do not certify a positive key. It does not demonstrate insecurity or establish an optimal loss limit.

Figure 4 displays convergence with block size at fixed losses of 0, 10 and 20 dB. For fixed loss and coefficients, both the honest score margin and the leading entropy-accumulation penalty per emission vanish as $n^{-1/2}$ when $\alpha - 1$ is proportional to $n^{-1/2}$. The verification terms and reconciliation rounding vanish as n^{-1}. Since the tangent touches the reference privacy term, the finite design converges to the same asymptotic rate, rather than to a different honest model or an arbitrarily reduced entropy threshold.

The large block requirements expose the cost of the global range and variance bounds used in the entropy-accumulation correction. They motivate tighter attainable-score variance estimates and improved parameter optimization, while preserving the present certified baseline. The numerical comparison is an evaluation of the derived formulas, not an experimental measurement or a simulation restricted to selected attacks. The hardware reduction and trust assumptions remain those of Sect. 5. Only the score-test contribution to honest

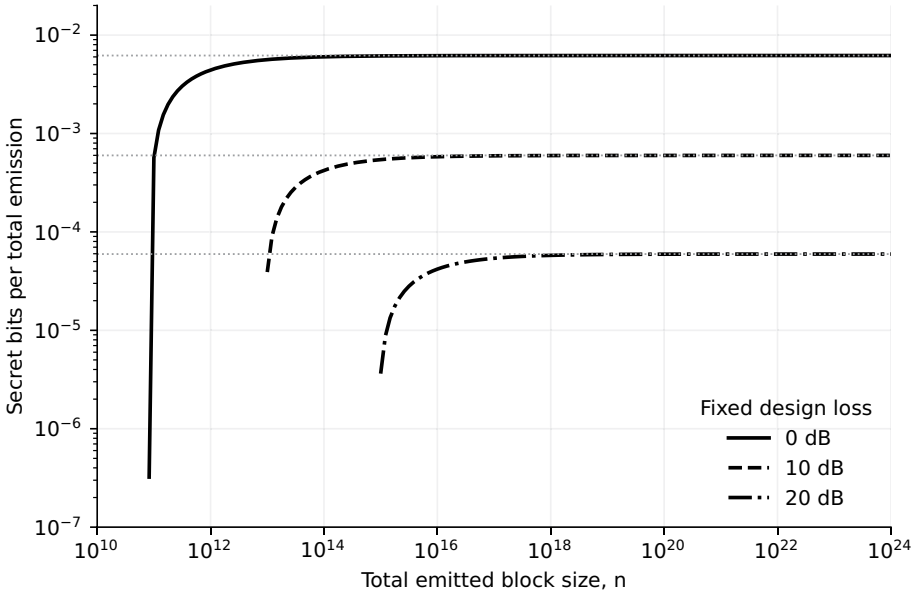


Fig. 4 Finite-block convergence at fixed nominal loss, with all other optical parameters unchanged. Gray dotted horizontal lines indicate the corresponding asymptotic rates. Nonpositive finite bounds are omitted from the logarithmic axis. Block size denotes emitted pulses, not accepted or sifted bits. The large blocks illustrate the scaling of the conservative bound and are not asserted to be practical acquisition targets

abort is bounded here. Total completeness additionally requires an implemented reconciliation and verification procedure with a quantified success probability.

7 Conclusion

This work established a channel-level and security-level description of distributed passive-user Loop-Back QKD. The direct random-rotation channel and the anisotropic dephasing representative reproduce the same BB84 statistics while differing on the unmeasured Y axis. Their comparison makes explicit what Alice's accessible data determine and what remains outside the operational equivalence class.

Security does not rely on either honest channel representative. Alice provides trusted local quantum-service infrastructure, while Eve is allowed an arbitrary adaptive coherent operation on the single interface between the two user rotations. A Choi-test identity gives the ideal relation $e_{1,\mathrm{p}} = 3e_{1,\mathrm{b}}/2$, and the calibrated angular factor κ_{rot} extends it to imperfect rotators. Together with phase randomization, decoy-state estimation and generalized entropy accumulation, this relation yields a finite-key composable bound.

The common-model numerical evaluation quantifies how finite blocks reduce the certified loss range and shows convergence to the asymptotic rate. Its large block requirements identify a limitation of the conservative correction, rather than an experimental operating claim.

The analysis also identifies the physical boundary of the result. Allowing Eve to surround a user rotator enables loss-assisted operation discrimination. Trusted placement of the source and receiver, optical isolation, a qubit reduction and characterized detector processing are therefore part of the protocol specification. Within these conditions, lightweight users can establish a key through local unitary actions on infrastructure supplied by Alice without requiring sources or detectors of their own.

Appendix A: Choi-operator proof of the phase-error relation

Let ρ be one of the four BB84 states and let M_ρ be the positive operator-valued measurement element representing Alice's orthogonal outcome. For a completely positive map $\mathcal{N}$, which may be a trace-nonincreasing branch of Eve's instrument, the non-normalized state of the virtual choice registers after acceptance has matrix elements

$$[\omega_{S_1S_2}]_{xy,x'y'} = \frac{1}{4}\,\mathrm{Tr}\big[M_\rho R_2(y)\mathcal{N}\big(R_1(x)\rho R_1(x')^\dagger\big)\,R_2(y')^\dagger\big]. \tag{74}$$

The uniform average over ρ is understood. Define

$$p_F = \mathrm{Tr}\,\omega, \qquad q_{1,\mathrm{b}} = \mathrm{Tr}(P_\mathrm{b}\omega), \qquad q_{1,\mathrm{p}} = \mathrm{Tr}(P_\mathrm{p}\omega). \tag{75}$$

Use the Choi convention

$$J_\mathcal{N} = \sum_{m,n} |m\rangle\langle n| \otimes \mathcal{N}(|m\rangle\langle n|), \tag{76}$$

for which

$$\mathrm{Tr}[M\mathcal{N}(A)] = \mathrm{Tr}[(A^\mathsf{T} \otimes M)J_\mathcal{N}]. \tag{77}$$

Substitution of (74) constructs test operators satisfying

$$p_F = \mathrm{Tr}(\Gamma_F J_\mathcal{N}), \quad q_{1,\mathrm{b}} = \mathrm{Tr}(\Gamma_\mathrm{b} J_\mathcal{N}), \quad q_{1,\mathrm{p}} = \mathrm{Tr}(\Gamma_\mathrm{p} J_\mathcal{N}). \tag{78}$$

The uniform BB84 ensemble satisfies

$$\frac{1}{4}\sum_j \rho_j^\mathsf{T} \otimes (I_2 - \rho_j) = \frac{1}{4}I_4 - \frac{1}{8}S, \qquad S = X \otimes X + Z \otimes Z. \tag{79}$$

Writing $C = \cos(2a_1)\cos(2a_2)$ and $D = \cos(2a_1 - 2a_2)$, averaging the diagonal choice terms yields

$$\Gamma_F = \frac{1}{4}I_4 - \frac{C}{8}S, \qquad \Gamma_\mathrm{b} = \frac{1}{8}I_4 - \frac{D}{16}S. \tag{80}$$

For the cross terms selected by $X \otimes X$, let $\rho_j = (I_2 + \boldsymbol{v}_j \cdot \boldsymbol{\sigma})/2$. Since $\boldsymbol{v}_j$ lies in the X-Z plane,

$$\frac{1}{2}\sum_x R_1(x)\rho_j R_1(1-x)^\dagger = \frac{1}{2}[\cos(2a_1)I_2 + \boldsymbol{v}_j \cdot \boldsymbol{\sigma}]. \tag{81}$$

The corresponding output cross term is $[\cos(2a_2)I_2 - \boldsymbol{v}_j \cdot \boldsymbol{\sigma}]/2$. Their ensemble average gives $\Gamma_{XX} = CI_4/4 - S/8$, and therefore

$$\Gamma_{\mathrm{p}} = \frac{\Gamma_F + \Gamma_{XX}}{2} = \frac{1+C}{8}I_4 - \frac{1+C}{16}S. \tag{82}$$

For $a_1 = a_2 = \pi/8$, $C = 1/2$ and $D = 1$, so the computational input-output matrices are

$$\Gamma_F = \frac{1}{16}\begin{pmatrix} 3 & 0 & 0 & -1 \\ 0 & 5 & -1 & 0 \\ 0 & -1 & 5 & 0 \\ -1 & 0 & 0 & 3 \end{pmatrix}, \tag{83}$$

and

$$\Gamma_{\mathrm{b}} = \frac{1}{16}\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 3 & -1 & 0 \\ 0 & -1 & 3 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}. \tag{84}$$

Evaluating P_{p} in the same expression yields the operator identity

$$\Gamma_{\mathrm{p}} = \frac{3}{2}\Gamma_{\mathrm{b}}. \tag{85}$$

Since every completely positive map has $J_{\mathcal{N}} \geq 0$, contraction of (85) with any admissible $J_{\mathcal{N}}$ proves Lemma 2. If Eve retains a quantum output, each tester is tensored with the identity on that output, so the identity and the conclusion are unchanged.

Appendix B: Robust relation for imperfect rotations

Let a_1, a_2 be the actual positive rotation magnitudes and define

$$C = \cos(2a_1)\cos(2a_2), \qquad D = \cos(2a_1 - 2a_2), \qquad r = \frac{D-1}{D-C}. \tag{86}$$

Substitution of $R_i(\pm a_i) = \cos a_i I \mp i \sin a_i Y$ into the three Choi testers gives, entry by entry,

$$\Gamma_{\mathrm{p}} = (1+C)\left[(1-r)\Gamma_{\mathrm{b}} + \frac{r}{2}\Gamma_F\right]. \tag{87}$$

Consequently,

$$e_{1,\mathrm{p}} = (1+C)\left[e_{1,\mathrm{b}} + \frac{D-1}{D-C}\left(\frac{1}{2} - e_{1,\mathrm{b}}\right)\right]. \tag{88}$$

For $a_i \in (0, \pi/4)$, $D \leq 1$ and $D - C = \sin(2a_1)\sin(2a_2) > 0$. In the low-error regime $e_{1,\mathrm{b}} \leq 1/2$, the second term inside the bracket is nonpositive. Hence

$$e_{1,\mathrm{p}} \leq (1+C)e_{1,\mathrm{b}}. \tag{89}$$

Maximizing C under $|a_i - \pi/8| \leq \Delta_i$ gives (57).

The entropy statement (58) follows from the uncertainty relation

$$H(Z_{S_1} \mid E, I_{\mathrm{pre}}, F, m=1) + H(X_{S_1} \mid S_2, I_{\mathrm{pre}}, F, m=1) \geq 1. \tag{90}$$

A virtual X-basis measurement of S_2, with the sign convention for $|\Phi^-\rangle$, has disagreement probability $e_{1,\mathrm{p}}$. The binary Fano inequality gives

$$H(X_{S_1} \mid S_2, I_{\mathrm{pre}}, F, m=1) \leq h_2(e_{1,\mathrm{p}}), \tag{91}$$

The uncertainty and Fano bounds are first applied conditional on each preparation. Concavity of binary entropy then bounds their acceptance-weighted average by the binary entropy of the averaged phase error used in the Choi identity. For $\kappa_{\mathrm{rot}} e_{1,\mathrm{b}} \geq 1/2$, nonnegativity of classical conditional entropy supplies the zero contribution. This completes the reduction without extrapolating the low-error angular inequality.

Appendix C: Affine tradeoff and explicit finite-key correction

Define

$$\phi_\kappa(e) = 1 - h_2\left(\min\left\{\frac{1}{2}, \kappa e\right\}\right). \tag{92}$$

For a chosen tangent point $0 < e_\star < 1/(2\kappa)$, convexity gives

$$y\phi_\kappa(z/y) \geq A_\star y + B_\star z, \tag{93}$$

where

$$B_\star = -\kappa \log_2 \frac{1-\kappa e_\star}{\kappa e_\star}, \qquad A_\star = 1 + \log_2(1-\kappa e_\star). \tag{94}$$

Here, $A_\star > 0$ and $B_\star < 0$. Write the decoy estimators as

$$Y^L_{1,F} = a_\nu Q_{\nu,F} + a_\mu Q_{\mu,F} + a_0 Q_{0,F}, \qquad Z^U_{1,\mathrm{b}} = b_\nu T_{\nu,F} + b_0 Q_{0,F}, \tag{95}$$

with

$$a_\nu = \frac{\mu e^\nu}{\mu\nu - \nu^2}, \quad a_\mu = -\frac{\nu^2 e^\mu}{\mu(\mu\nu - \nu^2)}, \quad a_0 = -\frac{\mu^2 - \nu^2}{\mu(\mu\nu - \nu^2)}, \tag{96}$$

$$b_\nu = \frac{e^\nu}{\nu}, \quad b_0 = -\frac{1}{2\nu}. \tag{97}$$

Let q be the per-emission distribution of the score symbols μF, νok, νerr, $0F$ and $\perp$, where $\perp$ collects all remaining events. The symbol μF counts accepted signal rounds before the test selection. The retained-key entropy therefore includes the factor $1-\tau$, while all revealed test bits belong to the public transcript. Thus

$$q_{\mu F} = p_\mu Q_{\mu,F}, \quad q_{\nu\mathrm{ok}} = p_\nu (Q_{\nu,F} - T_{\nu,F}), \tag{98}$$

$$q_{\nu\mathrm{err}} = p_\nu T_{\nu,F}, \quad q_{0F} = p_0 Q_{0,F}. \tag{99}$$

An affine min-tradeoff function is

$$f(q) = r_{\mu F} q_{\mu F} + r_{\nu\mathrm{ok}} q_{\nu\mathrm{ok}} + r_{\nu\mathrm{err}} q_{\nu\mathrm{err}} + r_{0F} q_{0F}, \tag{100}$$

where

$$r_{\mu F} = (1-\tau)\mu e^{-\mu} A_\star a_\mu, \tag{101}$$

$$r_{\nu\mathrm{ok}} = \frac{p_\mu (1-\tau)\mu e^{-\mu}}{p_\nu} A_\star a_\nu, \tag{102}$$

$$r_{\nu\mathrm{err}} = \frac{p_\mu (1-\tau)\mu e^{-\mu}}{p_\nu} (A_\star a_\nu + B_\star b_\nu), \tag{103}$$

$$r_{0F} = \frac{p_\mu (1-\tau)\mu e^{-\mu}}{p_0} (A_\star a_0 + B_\star b_0), \qquad r_\perp = 0. \tag{104}$$

Equation (93), the lower bound on $Y_{1,F}$, and the upper bound on $Z_{1,\mathrm{b}}$ imply that $f(q)$ lower-bounds the conditional entropy generated in one round. The signs of $A_\star$ and $B_\star$ preserve the inequality directions.

For a directly computable, conservative generalized-entropy-accumulation correction, define

$$f_{\max} = \max_c r_c, \quad f_{\min} = \min_c r_c, \quad R_f = f_{\max} - f_{\min}, \tag{105}$$

including $r_\perp = 0$, and set $d_K = 2$ for the padded binary output register defined before Theorem 3. Popoviciu's inequality gives $\mathrm{Var}(f) \le R_f^2/4$. Therefore, a valid upper bound on the variance-dependent quantity in the generalized entropy accumulation theorem is

$$V_{\mathrm{ub}}^2 = \left[\log_2(2d_K^2 + 1) + \sqrt{2 + \frac{R_f^2}{4}} \right]^2. \tag{106}$$

For $1 < \alpha < 3/2$, use

$$K_{\mathrm{ub}}(\alpha) = \frac{(2-\alpha)^3}{6(3-2\alpha)^3 \ln 2} 2^{\frac{\alpha-1}{2-\alpha}(2\log_2 d_K + R_f)} \times \ln^3\left(2^{2\log_2 d_K + R_f} + e^2\right). \tag{107}$$

For $0 < \varepsilon_{\mathrm{s}} < 1$, define

$$g(\varepsilon_{\mathrm{s}}) = -\log_2\left(1 - \sqrt{1-\varepsilon_{\mathrm{s}}^2}\right). \tag{108}$$

The correction in Theorem 3 is

$$\begin{aligned} &\Delta_{\mathrm{GEAT}}(n, \alpha, f, \varepsilon_{\mathrm{s}}, \varepsilon_{\mathrm{EA}}) \\ =& n\frac{\alpha-1}{2-\alpha}\frac{\ln 2}{2} V_{\mathrm{ub}}^2 + \frac{g(\varepsilon_{\mathrm{s}}) + \alpha \log_2(1/\varepsilon_{\mathrm{EA}})}{\alpha - 1} + n\left(\frac{\alpha-1}{2-\alpha}\right)^2 K_{\mathrm{ub}}(\alpha). \end{aligned} \tag{109}$$

This is a conservative specialization of the conditioned bound in Theorem 4.3 of Ref. Metger et al. (2024). For numerical evaluation, cancellation in the smoothing term is avoided by the equivalent expression $g(\varepsilon_{\mathrm{s}}) = -2\log_2 \varepsilon_{\mathrm{s}} + \log_2(1 + \sqrt{1-\varepsilon_{\mathrm{s}}^2})$.

To verify the sequential hypotheses, write $\mathsf{E}_i = E_i I^i$ for Eve's memory together with all accumulated public records. The round map is $\mathcal{M}_i : \mathsf{E}_{i-1} \to K_i C_i \mathsf{E}_i$, with fresh trusted settings generated internally. No private output of an earlier round is an active input to later rounds. Consequently $\mathcal{R}_i = \mathrm{Tr}_{K_i C_i} \circ \mathcal{M}_i$ is a channel from E_{i-1} alone. If inert private records R_i are retained, the condition becomes

$$\mathrm{Tr}_{K_i C_i R_i} \circ \mathcal{M}_i = \mathcal{R}_i \circ \mathrm{Tr}_{R_{i-1}}. \tag{110}$$

Each score C_i is a deterministic function of classical records preserved in I^i. Copying these orthogonal classical labels reconstructs each score without disturbance, and the complete sequence can still be reconstructed from I^n. This supplies the local and global score-reconstruction conditions.

For every input memory state and its purifying reference $\widetilde{E}_{i-1}$, the Choi identity and uncertainty relation apply with that reference included in Eve's side information. Conditional on emitted photon number, intensity indistinguishability gives the same decoy inequalities for this input. Other photon-number sectors contribute a nonnegative classical conditional entropy and may be assigned zero. Independent test selection leaves a secret signal contribution with weight $p_\mu(1-\tau)\mu e^{-\mu}$. The tangent construction therefore establishes

$$H(K_i \mid \mathsf{E}_i \widetilde{E}_{i-1}) \geq f(q) \tag{111}$$

for every compatible round output with score distribution q. The bound does not assume independent attacks or a classical purifying reference.

Let $\Omega = \Omega_{\mathrm{PE}}$. If $\Pr \Omega \geq \varepsilon_{\mathrm{EA}}$, the conditioned accumulation theorem gives

$$H_{\min}^{\varepsilon_{\mathrm{s}}}(K^n \mid \mathsf{E}_n)_{|\Omega} \geq nh - \Delta_{\mathrm{GEAT}}. \tag{112}$$

The replacement of the unknown $\Pr\Omega$ by $\varepsilon_{\mathrm{EA}}$ yields the correction above. The classical leakage chain rule subtracts L_{EC}, the verification hash length and the one-bit verification announcement. The verification seed is sampled after the reconciled strings are fixed. Two-universality bounds the probability of accepting unequal strings by $\varepsilon_{\mathrm{EV}}$. An independent privacy-amplification seed and the quantum leftover-hash lemma (Tomamichel et al. 2011) bound the secrecy distance by

$$2\varepsilon_{\mathrm{s}} + \frac{1}{2}2^{[\ell - H_{\min}^{\varepsilon_{\mathrm{s}}}(K^n|E_{\mathrm{post}})]/2}, \tag{113}$$

where E_{post} contains all accounted classical communication. Equation (64) makes this at most $2\varepsilon_{\mathrm{s}} + \varepsilon_{\mathrm{PA}}$. Extraction can be defined virtually in aborted branches and followed by erasure of their key. Since the abort flags are already included in the leakage accounting, this final selection preserves the subnormalized secrecy bound by contractivity of trace distance.

If $\Pr\Omega < \varepsilon_{\mathrm{EA}}$, the final key is produced with probability at most $\varepsilon_{\mathrm{EA}}$, which directly bounds its subnormalized secrecy distance. Combining the two cases conservatively proves (65). Thus the event used in accumulation is a statistical event before reconciliation, while Ω_{fin} denotes final non-abort.

The region $\mathcal{S}_{\mathrm{acc}}$ is fixed directly on empirical scores. For affine constraints, its entropy minimum is obtained by a linear program. These empirical frequencies need not equal an underlying stationary channel distribution. Additional confidence procedures for adaptive conditional means require concentration inequalities valid for adapted sequences and a justified conversion of their failure event into the secrecy criterion. Exact binomial intervals require an independently justified independent-trial model. The present theorem does not add an unspecified estimation-failure term. Completeness is separate: for a specified honest model, the probability of failing the tests or classical processing is bounded by a declared honest-abort parameter.

Appendix D: Security budget and reproducibility checklist

A reproducible evaluation specifies the emitted block size n, photon-number variable m , nominal intensities and their calibrated validity, intensity probabilities, test probability τ, angular bounds Δ_1, Δ_2, score counts, the predetermined empirical acceptance region, reconciliation alphabet budget L_{EC}, verification length and its announcement bit, the fixed output length ℓ, tangent $e_\star$, order α, and all declared failure parameters. The parameters $\varepsilon_{\mathrm{s}}, \varepsilon_{\mathrm{EA}}, \varepsilon_{\mathrm{PA}}, \varepsilon_{\mathrm{EV}}$ are fixed before acquisition. The total ideal-authentication parameter is $\varepsilon_{\mathrm{sec}} + \varepsilon_{\mathrm{EV}}$. Nonideal authentication adds its failure probability, and consumed authentication key is accounted for separately in the net yield.

Reconciliation messages may be padded to a fixed maximum length. Any data-dependent length, interactive message or branch label must fit within the declared alphabet budget. Exceeding that budget causes abort. This fixed-length statement does not claim an unproved adaptive-length improvement based solely on the observed syndrome size.

Hardware use additionally requires phase randomization, intensity indistinguishability, the common receiver reduction in (50) or an explicit optical proof, specified double-click processing, characterized efficiency mismatch, and protection against spectral, temporal and

injected-probe leakage. Intensity uncertainty requires bounds valid over the entire calibrated range. The nominal-intensity formulas are not automatically extended by substituting favorable interval endpoints. These conditions define the interface to which the theorem applies.

Honest score-test probability for the numerical design

For the independent honest model only, let $Z_i = r_{C_i}$, so that $\mathbb{E}Z_i = f(q^{\mathrm{ref}})$ and Z_i lies in an interval of width R_f. The centered logarithmic moment-generating function has value and derivative zero at the origin. Its second derivative is the variance under exponential tilting, bounded by $R_f^2/4$ because the tilted distribution has the same support interval. Integrating twice gives

$$\mathbb{E}e^{-t(Z_i-\mathbb{E}Z_i)} \le e^{t^2R_f^2/8}, \qquad t \ge 0.$$

Independence, Markov's inequality and optimization at $t = 4m/R_f^2$ yield

$$\Pr\left[\frac{1}{n}\sum_i Z_i - f(q^{\mathrm{ref}}) \le -m\right] \le \exp(-2nm^2/R_f^2).$$

Substitution of $m = m_n$ proves the score-test statement in Sect. 6.2. This argument concerns honest completeness of that test. The adversarial secrecy proof continues to use the empirical acceptance region and the coherent-attack theorem, without an independence assumption on Eve.

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Data availability No experimental data were collected. The illustrative numerical calculations use the parameters and models stated in the manuscript.

Materials availability Not applicable.

Code availability Verification scripts provided with the revision reproduce the channel identities, the operator certificate, and the illustrative rate calculations.

Declarations

Conflict of interest The author declares no conflicts of interest.

Ethical approval Not applicable.

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